

SUMMARY

Software Engineer with **production experience** building **backend services, REST APIs and high-volume applications** using **Java/Spring Boot, Python, Go/Rust and AWS**. Skilled in application architecture, solution design, microservices, DSA, automated testing, CI/CD, Kubernetes, SQL databases, observability, security and role-based business rules in enterprise systems.

TECHNICAL SKILLS

Languages: Java 8+, Python, SQL, JavaScript, TypeScript, Go, Rust, C++ — **Backend:** Spring Boot, Django, Django REST Framework, FastAPI, Node.js, Express.js, REST APIs, Microservices — **Databases:** PostgreSQL, TimescaleDB, Redis, MongoDB, MySQL, Oracle, Firebase Firestore — **Cloud & Infrastructure:** AWS, EC2, S3, Docker, Kubernetes, GitHub Actions, Jenkins, Maven, Nginx, CI/CD — **Testing:** JUnit, Mockito, pytest, Selenium WebDriver, Unit Testing, Integration Testing, Functional Testing, API Testing, Security Testing, Mocking — **Engineering:** Application Architecture, API Design, DSA, Design Patterns, Business Rules, Functional Requirements, Non-Functional Requirements, Performance Optimization, Authentication, Authorization, RBAC, Production Troubleshooting — **Observability:** Grafana, Plutono, Splunk, Monitoring, Time-Series Analytics, Production Observability, APM — **Development:** Git, GitHub, Source Code Management (SCM), Version Control, Jira, Confluence, Agile Methodology, Scrum, Scripting — **Frontend:** React, React Native, Next.js, Angular

EXPERIENCE

Full Stack Engineer

Feb 2026 - Present

Zwilling Labs (IIT Bombay)

- Owned backend development across **8 enterprise deployments**, building numerous **Go/Rust services and REST APIs** for high-volume, **high-performance big-data and industrial telemetry workloads** on a customized Plutono/Grafana stack.
- Owned the **Hitachi deployment end-to-end**, supporting **150+ machines** with operator, supervisor and admin workflows spanning identification, machine assignment, task allocation, start/pause/resume/stop operations, pause-reason capture and **role-based business rules**.
- Designed **application architecture, API contracts, service orchestration and end-to-end data flows** across scanner, frontend, backend and database layers, translating functional and **non-functional requirements** into production services; built PostgreSQL/TimescaleDB schemas, relationships and indexes for telemetry and operational workloads.
- Reduced automated PDF report generation from **hours to under 2 minutes** by building Python-based reporting workflows and Chromium-based rendering for production reporting.
- Deployed production services using **Docker/Kubernetes**, including Services, ConfigMaps, Secrets and scaling; built **GitHub Actions CI/CD** workflows across private-cloud, edge and on-premise environments.
- Performed **unit, integration, functional, API and security testing**, including boundary, failure and worst-case scenarios; used **production monitoring, observability and Splunk-based log analysis** for troubleshooting and performance optimization while **mentoring 2 interns**.
- Worked in **Agile Methodology/Scrum** using sprints, stand-ups, retrospectives, Jira and Confluence while managing multiple enterprise development and deployment workflows within project timelines.

Software Engineer (Backend)

Jan 2025 - Jan 2026

Vervelo

- Built backend systems across **4+ client projects** using **Python/Django and Java/Spring Boot**, developing REST APIs for healthcare workflows spanning profiles, appointments, notifications, messaging, reports and payment integrations.
- Reduced API response time by approximately **30%** through SQL query optimization, indexing and **Redis caching**; implemented authentication, authorization, RBAC, background processing and production database workflows.
- Performed comprehensive **unit, integration, functional and API testing** across normal, boundary and failure-path scenarios using **JUnit/pytest**; worked with **Maven, Jenkins, Git/GitHub and Dockerized services** across CI/CD workflows.
- Deployed backend services on **AWS EC2**, performed production root-cause analysis across application/API/database layers and collaborated with frontend/UI teams to deliver production features.

Software Engineer Intern

Feb 2024 - Aug 2024

Thinkitive Technologies, Pune

- Built **React enterprise dashboards**, integrated REST APIs and implemented reusable components and frontend-backend data flows; performed API/integration testing and debugging.

RESEARCH EXPERIENCE

Research Project - Lind-Wasm

2026 - Present

with *Justin Cappos, New York University* — Contributing to **Lind-Wasm**, an OS/security research system providing **software fault isolation** for POSIX-like applications in WebAssembly runtimes; working with the **Rust** codebase, **Wasmtime**, modified glibc and system-call mediation through **3i**, with components for file-descriptor management, virtual-memory mapping, signal handling and WebAssembly optimization via **Binaryen**.

Research Project - CodeRevival

2026 - Present

with *Dennis Shasha, New York University* — Developing **CodeRevival**, a system for automatically reviving older GitHub repositories by discovering outdated software dependencies and test setups and identifying newer dependency configurations that preserve successful execution; working on **VersionClimber**, which searches package-version configurations to migrate legacy repositories toward newer dependencies while maintaining compatibility with existing code and tests.

SELECTED PRODUCT BUILDS

Mesh - Multi-Agency SaaS Operating System

[Demo](#)

- Built and deployed a configurable **multi-agency SaaS platform** used by **200+ active users** with **7 core modules** spanning sales pipelines, deal tracking, employees, attendance, leave, tasks and performance management; designed **REST APIs, relational data models, authentication, RBAC, role-based permissions, workflow states and operational reporting** across configurable agency operations.

Hiuen - Live Social Platform

[Demo](#)

- Built and deployed a live social matching platform used by **150+ users**, supporting authentication, profiles, discovery, matching, chat and real-time interactions using **React Native, Node.js, Express.js and MongoDB**; designed **REST APIs, asynchronous data flows, MongoDB schemas and pagination** and deployed backend infrastructure on **AWS EC2/S3**.

Morph - Multi-Restaurant Digital Menu SaaS

[Demo](#)

- Built a configurable **multi-restaurant SaaS platform** covering digital menus, QR ordering, customer onboarding, feedback and review workflows, including a **canvas-style visual layout editor**; developed end-to-end **WhatsApp Business/Meta API integration** for campaign messaging, feedback notifications and automated customer communication.

EDUCATION

Fr. Conceicao Rodrigues College of Engineering, Mumbai

Nov 2021 - Mar 2025

B.E. Computer Science (Honors in Blockchain)